

Practical Applications of I4C and OSINT in Cybersecurity Investigation and Threat Intelligence

Practical Task 1

Analyze Daily Cyber Threats Using I4C Daily Digest

Task Definition

Access the I4C Daily Digest portal to identify the latest cybercrime incidents, attack trends, and security advisories. Prepare a summary highlighting the attack type, targeted sector, and recommended preventive measures.

Steps

1. Open the I4C Daily Digest portal.
2. Read the latest published cybercrime reports.
3. Identify at least three recent cyber incidents.
4. Record:
 - Type of cyber attack
 - Targeted organization/sector
 - Impact
 - Prevention techniques
5. Prepare a one-page summary.
6. Discuss how the attacks could have been prevented.

Practical Task 2

Identify Cyber Threat Categories from I4C

Task Definition

Study different categories of cybercrime reported by I4C and classify them based on attack objectives and victim profiles.

Steps

1. Visit the I4C portal.
2. Explore different cybercrime categories.
3. Create a table containing:
 - Financial Fraud

- Social Media Crime
 - Identity Theft
 - Ransomware
 - Phishing
 - OTP Fraud
4. Mention the attack mechanism.
 5. List preventive security practices.
 6. Submit the classification report.

Practical Task 3

Explore the OSINT Framework

Task Definition

Understand the structure of the OSINT Framework by identifying various categories of publicly available intelligence sources used in cybersecurity investigations.

Steps

1. Open the OSINT Framework website.
2. Explore major investigation categories.
3. Select five OSINT categories.
4. Record:
 - Purpose
 - Available tools
 - Information collected
5. Prepare a comparison table.
6. Discuss ethical use of OSINT.

Practical Task 4

Email Investigation Using OSINT Tools

Task Definition

Perform ethical analysis of a publicly available sample email address using OSINT resources to identify publicly accessible information without violating privacy.

Steps

1. Open the OSINT Framework.
2. Navigate to Email Investigation tools.
3. Select one publicly available sample email.
4. Perform reputation/search lookup.
5. Record publicly available findings.
6. Document limitations and privacy considerations.
7. Prepare an investigation report.

Practical Task 5

Domain Information Collection Using OSINT

Task Definition

Collect publicly available information about a selected organization's domain to understand digital footprints.

Steps

1. Open the OSINT Framework.
2. Select Domain Investigation tools.
3. Choose an educational or public domain.
4. Collect:
 - WHOIS information
 - DNS records
 - IP address
 - Hosting details
5. Record observations.
6. Explain how attackers and defenders use such information.

Practical Task 6

Website Technology Identification

Task Definition

Identify technologies used by a public website using OSINT techniques and analyze the technology stack from a security perspective.

Steps

1. Select a public website.
2. Use Website Analysis tools listed in the OSINT Framework.
3. Identify:
 - Web server
 - CMS
 - Programming language
 - Framework
 - SSL availability
4. Record findings.
5. Suggest possible security best practices.

Practical Task 7

Social Media Intelligence Collection

Task Definition

Collect publicly available organizational information from official social media accounts using ethical OSINT practices.

Steps

1. Open the OSINT Framework.
2. Select Social Network Investigation.
3. Choose an organization's official profile.
4. Collect:
 - Organization information
 - Public posts
 - Official announcements
 - Security awareness posts
5. Record observations.
6. Discuss digital footprint management.

Practical Task 8

Cyber Threat Intelligence Report Preparation

Task Definition

Prepare a Cyber Threat Intelligence (CTI) report using information collected from I4C and OSINT resources.

Steps

1. Read the latest I4C Daily Digest.
2. Select one cyber incident.
3. Search related public information using the OSINT Framework.
4. Identify:
 - Threat actor (if publicly reported)
 - Attack vector
 - Impact
 - Indicators
 - Mitigation techniques
5. Prepare a structured CTI report.
6. Present findings.

Practical Task 9

Security Awareness Poster Based on I4C Advisories

Task Definition

Design a cybersecurity awareness poster using recommendations published by I4C.

Steps

1. Read recent I4C advisories.
2. Select one cybercrime topic.
3. Identify:
 - Warning signs
 - Prevention methods
 - Reporting procedure
4. Design an awareness poster.
5. Include emergency reporting information.
6. Present the poster in class.

Practical Task 10

Comparative Study of Encryption and OSINT in Cybersecurity

Task Definition

Analyze how encryption techniques (Caesar Cipher, AES, RSA) and OSINT contribute to modern cybersecurity operations.

Steps

1. Review the practical implementations of:
 - Caesar Cipher
 - AES
 - RSA
2. Explore the OSINT Framework.
3. Compare:
 - Purpose
 - Security objective
 - Real-world applications
 - Advantages
 - Limitations
4. Prepare a comparative analysis table.
5. Write conclusions on the role of encryption and OSINT in cyber defense.